

well-drained, neutral to acidic soil. Water regularly, less when not in full growth. Stems must be tied to supports. Prune previous season's growth back to 1 or 2 nodes in spring. Propagate by softwood cuttings in spring or summer. Whitefly and red spider mite may be troublesome. ☉Contact with sap may irritate skin; all parts may cause mild stomach upset if ingested.
A. cathartica (Golden trumpet). **‘Hendersonii’** illus. p.464.

ALLIUM

Onion

ALLIACEAE/LILIACEAE

Genus of perennials, some of which are edible, with bulbs, rhizomes or fibrous rootstocks. Nearly all have narrow, basal leaves smelling of onions when crushed, and most have small flowers packed together in a dense, spherical or shuttlecock-shaped umbel. Dried umbels of tall border species are good f or winter decoration. Requires an open, sunny situation and well-drained soil; best left undisturbed to form clumps. Plant in fall. Propagate by seed in fall or by division of clumps—spring-flowering varieties in late summer and summer-flowering ones in spring. ☉Contact with the bulbs may irritate skin or aggravate skin allergies.

A. acuminatum, syn. *A. murrayanum*, illus. p.416.
A. aflatunense illus. p.382.
A. aflatunense of gardens. See *A. x hollandicum*.
A. akaka illus. p.416.
A. albobulosum. See *A. cristophii*.
A. atropurpureum illus. p.392.
A. azureum. See *A. caeruleum*.
A. beesianum. Clump-forming, late-summer-flowering bulb. H to 8in (20cm), S 2in (5cm). Has linear, gray-green leaves and bears pendent heads of bell-shaped, blue flowers. Z6–10
A. caeruleum, syn. *A. azureum*, illus. p.411.
A. carinatum subsp. pulchellum, syn. *A. pulchellum*. Clump-forming, summer-flowering bulb. H 1–2ft (30–60cm), S 2in (5cm). Linear, semierect leaves sheathe lower two-thirds of stem. Has an umbel of pendent, cup-shaped, purple flowers. Z6–9
A. cernuum illus. p.409.
A. christophii. See *A. cristophii*.
A. cowanii. See *A. neapolitanum*.
A. cristophii, syn. *A. albobulosum*, illus. p.411.
A. cyaneum. Tuft-forming, summer-flowering bulb. H 4–10in (10–25cm), S 3in (8cm). Leaves are threadlike and erect. Stems each bear a small, dense umbel of 5 or more pendent, cup-shaped, blue or violet-blue flowers. ¼in (0.5cm) long. Z5–9
A. cyathophorum var. farreri illus. p.423.
A. flavum illus. p.412.
A. giganteum illus. p.392.
A. ‘Gladiator’ illus. p.392.
A. ‘Globemaster’ illus. p.392.
A. x hollandicum, syn. *A. aflatunense* of gardens. Tuft-forming, summer-flowering bulb. H 3ft (1m), S 4in (10cm). Has mid-green, basal leaves dying away by flowering time. Bears numerous

star-shaped, purplish-pink flowers in a dense, spherical umbel, 4in (10cm) across. Z4–11
A. insubricum. Clump-forming, summer-flowering bulb. H 6–10in (15–25cm), S 2in (5cm). Has narrowly strap-shaped, mid-green, stem-sheathing leaves. Bears umbels of 3–5 pendent, bell-shaped, pink-purple flowers, followed by pendent seed heads. Often confused with *A. narcissiflorum*, which has more flowers and erect seed heads. Z7–9
A. kansuense. See *A. sikkimense*.
A. karataviense illus. p.416.
A. moly illus. p.424.
A. ‘Mount Everest’ illus. p.385.
A. multibulbosum. See *A. nigrum*.
A. murrayanum. See *A. acuminatum*.
A. murrayanum of gardens. See *A. unifolium*.
A. narcissiflorum, syn. *A. pedemontanum* of gardens, illus. p.422.
A. neapolitanum, syn. *A. cowanii*, illus. p.399.
Cowanii Group illus. p.409.
A. neriniflorum, syn. *Caloscordum neriniflorum*, *Nothoscordum neriniflorum*. Clump-forming, late summer-flowering bulb. H 4–10in (10–25cm), S 3–4in (8–10cm). Threadlike, semierect, basal leaves die down at flowering time. Each leafless stem bears a loose head of 10–20 small, funnel-shaped, pinkish-red flowers. Z3–8
A. nigrum, syn. *A. multibulbosum*. Clump-forming, summer-flowering bulb. H 14in (35cm), S 3in (8cm). Produces flattish umbels of 20–35 large, open cup-shaped flowers, usually creamy white, sometimes pale lilac, each with a prominent dark green ovary. Basal leaves are lance-shaped, gray-green. Z7–9
A. oreophilum, syn. *A. ostrowskianum*, illus. p.418.
A. ostrowskianum. See *A. oreophilum*.
A. pedemontanum of gardens. See *A. narcissiflorum*.
A. pulchellum. See *A. carinatum* subsp. *pulchellum*.
A. ‘Purple Sensation’ illus. p.392.
A. rosenbachianum. Tuft-forming, summer-flowering bulb. H 3ft (1m), S 4in (10cm). Has ridged stems and gray-green straplike, basal leaves. Produces 50 or more star-shaped, deep purple flowers in a spherical umbel, 4in (10cm) across. Z6–9
A. schoenoprasum (Chives) illus. p.422.
A. schubertii illus. p.409.
A. senescens var. calcareum. See *A.s. subsp. montanum*. **subsp. montanum** (syn. *A.s. var. calcareum*) illus. p.409.
A. sikkimense, syn. *A. kansuense*. Tuft-forming, summer-flowering bulb. H 6–10in (15–25cm), S 4in (10cm). Leaves are linear, erect and basal. Up to 15 bell-shaped, blue flowers are borne in a small, pendent umbel, 1in (25cm) across. Z6–10
A. sphaerocephalon. Clump-forming, summer-flowering bulb. H 20–36in (50–90cm), S 3in (8cm). Has linear, semierect leaves on basal third of slender, wiry stems and a very dense umbel, ¾–1½in (2–4cm) across, of up to 40 small, bell-shaped, pinkish-purple flowers. Z4–11
A. stipitatum. Summer-flowering bulb. H to 3–4ft (1–1.5m), S 4in (10cm). Sturdy stems with straplike, semierect, basal leaves carry 50 or more star-shaped, purplish-pink flowers in a spherical umbel, 3–5in (8–12cm) across. Z4–9

A. tuberosum (Chinese chives). Clump-forming, late summer-to fall-flowering bulb. H 10–20in (25–50cm), S 2in (5cm). Has solid, linear, edible, basal and stem-sheathing leaves and star-shaped, fragrant white flowers borne in umbels. Z7–9
A. unifolium, syn. *A. murrayanum* of gardens, illus. p.402.

ALNUS

Alder

BETULACEAE

Genus of deciduous trees and shrubs, grown mainly for their ability to thrive in wet situations. Flowers are borne in catkins in late winter or early spring, the males conspicuous and attractive, the females forming persistent, woody, conelike fruits. Most do best in sun and any moist or even waterlogged soil, but *A. cordata* also grows well on poor, dry soils. Propagate species by seed sown in fall, cultivars by budding in late summer or by hardwood cuttings taken in early winter.
A. cordata (Italian alder). Fast-growing, deciduous, conical tree. H 80ft (25m), S 20ft (6m). Yellow, male catkins are produced in late winter and early spring, followed by heart-shaped, glossy, deep green leaves. Has persistent, round, woody fruits in fall. Z5–7
A. glutinosa (Black alder, Common alder). **‘Aurea’** is a slow-growing, deciduous, conical tree. H 40ft (12m), S 15ft (5m). Has rounded leaves, bright yellow until midsummer, later becoming pale green. Produces yellow-brown catkins in early spring. Useful grown in a boggy area. **‘Imperialis’** illus. p.75.
A. incana (Gray alder) illus. p.61. **‘Aurea’** (syn. *A. i.* ‘Ramulis Coccineis’) is a deciduous, conical tree. H 30ft (10m), S 15ft (5m). Has broadly ovate, yellow leaves, pale green in summer, and orange shoots and catkins in winter. Useful for cold, wet areas and poor soils. **‘Ramulis Coccineis’** see *A. i.* ‘Aurea’. Z2–6

ALOCASIA

ARACEAE

Genus of seasonally deciduous or evergreen perennials, usually with underground rhizomes, grown for their attractive foliage. Produces tiny flowers on a spadix enclosed in a leaflike spathe. Needs high humidity, partial shade and well-drained soil. Propagate by seed, stem cuttings or division of rhizomes in spring. ☉Contact with sap may irritate skin; all parts may cause mild stomach upset if ingested.
A. x amazonica. Rhizomatous perennial. H and S 3ft (1m). Has peltate, narrowly triangular, dark green leaves, to 2ft (60cm) long, with wavy, silvery-white margins, and leaf stalks to 2ft (60cm) long. Upper leaf surfaces have a metallic sheen and silvery-green veins; undersides are purple. Z9–12 (min. 59–64°F/15–18°C)
A. cuprea illus. p.472.
A. longiloba, syn. *A. lowii* var. *picta*, *A.l. var. veitchii*, *A. veitchii*. Very variable, evergreen, tufted perennial. H 2–5ft (60–150cm), S to 3ft (1m). Narrow leaves, triangular with arrow-shaped bases, are

ALOE

18in (45cm) long and green with grayish midribs, veins and margins, purple below. Has greenish spathes. Z14–15. (min. 59–64°F/15–18°C)
A. lowii var. *picta*. See *A. longiloba*.
var. veitchii see *A. longiloba*.
A. macrorrhiza (Giant elephant’s ear, Taro). Evergreen, tufted perennial with a thick, trunklike stem. H 12ft (4m), S 6–8ft (2–2.5m). Stalks 3ft (1m) long bear broad, arrow-shaped, glossy, green leaves, to 3ft (1m) long. Yellowish-green spathes are 8in (20cm) high. Z11. (min. 59–64°F/15–18°C)
A. veitchii. See *A. longiloba*.

ALOE

ASPHODELACEAE/LILIACEAE

Genus of evergreen, rosetted perennials. Some are shrublike or climbing, a few treelike, with succulent foliage and tubular to bell-shaped flowers. Tree aloes and shrubs with a spread over 1ft (30cm) prefer full sun; most smaller species prefer partial shade. Needs very well-drained soil. Propagate by seed, stem cuttings or offsets in spring or summer.
A. arborescens. Evergreen, treelike, many branched succulent. H 6–12ft (2–4m), S 6ft (2m). Stems are crowned by rosettes of widely spreading, long, slender, curved, dull blue-green leaves with toothed margins. Long flower stems bear abundant tubular to bell-shaped, red flowers 1½in (4cm) long in late winter and spring. Z11. (min. 50°F/10°C) **‘Variegata’** (illus. p.493) has numerous spikes of red flowers in late winter and spring.
A. barbadensis. See *A. vera*.
A. brevifolia. Basal-rosetted succulent producing many offsets. H 4in (10cm), S indefinite. Has broadly sword-shaped, fleshy, blue-green leaves with a few teeth along edges. In spring, flower stems, 20in (50cm) long, carry narrowly bell-shaped, bright red flowers 1¼–1½in (3–4cm) long. Z10–11 (min. 41°F/5°C)
A. broomii. Basal-rosetted succulent. H 4in (10cm), S 12in (30cm). Ovate to lance-shaped, mid-green leaves have minutely toothed, red-brown margins and sharp, terminal spines. In summer, unbranched flower stems, to 3ft (1m) long, bear dense spikes of tubular, pale lemon flowers, ¾in (2cm) long. Z10–11 (min. 41°F/5°C)
A. ciliaris. See *Aloiampelos ciliaris*.
A. concinna. See *A. squarrosa*.
A. descoingsii. Clump-forming, basal-rosetted succulent. H 1½–2in (4–5cm), S indefinite. Triangular, dull green leaves have white-toothed margins and many raised, white spots. In summer, unbranched flower stems, 6in (15cm) long, produce urn-shaped, red-orange flowers, ¼in (7mm) long. Z13–15 (min. 50°F/10°C)
A. distans. Trailing succulent. H 20in (50cm), S indefinite. Has stems often 6–10ft (2–3m) long, which root at the nodes. Lance-shaped to broadly ovate, fleshy, white-spotted, bluish-green leaves, 3in (8cm) long, are sharply pointed and have horny, yellow marginal teeth. In winter, panicles, 16–24in (40–60cm) long, with 3 or 4 branches, bear cylindrical, red and yellow flowers, 1½in (4cm) long, in terminal clusters. Z9–12
A. erinacea. See *A. melanacantha* var. *erinacea*.